INCH-POUND

MIL-DTL-32093A (OS)
22 February 2005
SUPERSEDING
MIL-DTL-32093 (OS)
3 October 2002

DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED
JAU-75/A, JAU-76/A, JAU-77/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the JAU-75/A, JAU-76/A, JAU-77/A Cartridge Actuated Initiators referred to herein as "initiator".

1.2 Application. These initiators are used to provide a time-delay function in various aircraft emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1377

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SPECIFICATIONS

FEDERAL

FED-STD-595	Colors used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-DTL-25579	Hose Assembly, Polytetrafluoroethylene, High Temperature, Medium Pressure, General Specification For

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	Environmental Engineering Considerations And Laboratory Tests
MIL-STD-1168	Ammunition Lot Numbering And Ammunition Data Card

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DOCUMENTS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

CAD-10003	Design Certification for the Initiator, Cartridge Actuated JAU-67/A, JAU-68/A, JAU-75/A, JAU-76/A, JAU-77/A

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn.: Engineering Documentation Branch (Code 4230), Indian Head, MD 20640-5035.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

298AS100	Closed Chamber (0.5 cubic inch)

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547AS100	Initiator, Cartridge Actuated (Delay)
726AS802	Fixture, Test
726AS804	Chamber Assembly, Closed (5.0 cubic inch)
848AS225	Initiator, Cartridge Actuated
3205AS266	Initiator, Cartridge Actuated JAU-76/A
3205AS267	Initiator, Cartridge Actuated JAU-75/A
3205AS268	Initiator, Cartridge Actuated JAU-77/A

(Application for copies should be addressed to the Commanding Officer, Naval Inventory Control Point, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DOD adopted, are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

MCCORMICK-SELPH INCORPORATED DRAWINGS

06331-826852	Initiator, Cartridge Actuated

(Application for copies of McCormick-Selph Inc. drawings should be sent to McCormick-Selph Inc. 3601 Union Rd., Hollister, CA 95024-0006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742	Standard Practice for Radiographic Inspection

(Copies of this document are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AS604	Hose Assembly, Polytetrafluoroethylene, Metallic Reinforced, 3000 psi, 400 Degrees F, Heavyweight, Hydraulic

(Copies of this document are available online at http://www.sae.org/ or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania, United States, 15096-001.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subjected to the first article inspection as specified in sections 4.2 and 6.3. A first article inspection shall be required if:

a. Two years or more have passed since the initiators were manufactured,
b. There has been a major turnover in personnel,
c. There have been major changes to the production facility or
d. There has been a transfer of production to a new facility or site.

3.2 Flight critical part. These initiators are flight critical items (see 6.4.1) installed on ejection seats and in escape systems and the technical data package shall be identified and procured as such.

3.3 Design and construction. The following applies to the qualification status and configuration control of the initiators over the entire life of the item.

3.3.1 Identical configuration. The initiator's design and construction shall be identical to that subjected to qualification testing and approved for service use by the Navy. Vendors desiring to supply these items shall meet the qualification requirements defined in CAD-10003. Changes to the configuration shall be verified by qualification testing. The qualified configurations are listed in Table I.

TABLE I. Configuration identification

Model No.	NAVAIR P/N	McCormick-Selph Inc
JAU-76/A	3205AS266	826852-02
JAU-75/A	3205AS267	826852-03
JAU-77/A	3205AS268	826852-04

3.3.2 Configuration control. Only the initiators procured from the vendor(s) listed on drawings 3205AS266, 3205AS267 and 3205AS268 are approved for the application specified. A substitute initiator shall not be used or procured without prior approval and testing by the Indian Head Division, Naval Surface Warfare Center, and the Naval Air Systems Command Headquarters.

3.4 Primary components. The percussion primers, propellants, Small Column Insulated Delay (SCID), and shear pins are primary components. Only one lot of each primary component shall be used in an initiator lot. One primary lot may be used in more than one initiator lot.

3.4.1 Percussion primers (primers). The primer shall be the following configuration: M42C1-979. Only one lot of each primary component shall be used in an initiator lot. One primary lot may be used in more than one initiator lot. Only primers manufactured within the last 24 months of assembly or re-certified within the last 12 months shall be used.

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3.4.2 Propellants. Only propellant manufactured within the last 24 months or re-certified within the last 12 months shall be used.

3.4.3 Small Column Insulated Delay (SCID). Only SCIDs from one lot manufactured within the last 24 months or re-certified within the last 12 months shall be used in the initiators.

3.5 Initiator production. The initiators shall be manufactured in accordance with the McCormick-Selph Inc. drawing 826852 and the drawings listed thereon. Each production initiator shall meet the requirements of 3.4 and 3.5.1 through 3.5.3 as outlined in Table II. Failure of any initiator to meet the requirements of 3.4 and Table II shall result in rejection of the initiator.

TABLE II. Production inspections.

Test Sequence	Test Paragraph	Requirement Paragraph
1. Visual	4.5.1	3.5.1, 3.5, 3.7
2. Radiographic X-ray Inspection	4.5.2	3.5.2
3. Radiographic N-ray Inspection	4.5.2	3.5.2
4. Leakage	4.5.3	3.5.3

3.5.1 Visual inspection. Per section 4.5.1 the initiators shall be free of the following defects: illegible, missing or inaccurate markings, damage, burrs, dents, sharp edges, bent or damaged mounting provisions, chipped or incomplete protective finish, or other defects which may prevent the installation of the initiators. The initiator shall have protective caps installed on the inlet and outlet ports. Each initiator shall meet the requirements of the drawings 3205AS266, 3205AS267 and 3205AS268 as required.

3.5.2 Radiographic examination. When radiographically examined in accordance with 4.5.2, the initiators shall show proper assembly, presence of all parts, and uniformity of the propellant and SCID column. The initiators shall be x-rayed such that the shear pin is visible along its longitudinal axis on the film. The radiographic inspection shall show no foreign materials present. Cartridges shall be subjected to N-ray inspection while assembled initiators shall be subjected to x-ray inspection.

3.5.3 Leakage. The leak rate of each initiator cartridge shall not exceed 1.0×10^{-5} cc/sec of air when tested in accordance with 4.5.3 prior to assembly into the initiator.

3.6 Initiator performance. The initiators shall meet all requirements of 3.6.1 through 3.7 and the acceptance criteria of 4.6.

3.6.1 Temperature-shock/humidity/altitude (TSHA) cycling. The initiators shall not initiate or incur any internal/external damage or degradation when subjected to TSHA cycling in accordance with 4.5.4, and shall meet the requirements of 3.5.1, 3.5.2 and 3.6.5 following the completion of the sequential environmental testing.

3.6.2 Vibration. The initiators shall not initiate or incur any internal/external damage or degradation when vibration cycles are applied in accordance with 4.5.5, shall be safe to handle, and shall meet the requirements of 3.5.1, 3.5.2 and 3.6.5 following completion of the sequential environmental testing.

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3.6.3 Shock. The initiators shall not initiate or incur any internal/external damage or degradation when sawtooth shocks are applied in accordance with 4.5.6, and shall meet the requirements of 3.5.1, 3.5.2 and 3.6.5 following completion of the sequential environmental testing.

3.6.4 6 Foot Drop. The initiators shall not initiate or incur any internal/external damage or degradation to performance when subjected to the 6-foot drop test in accordance with 4.5.7, and shall meet the requirements of 3.5.1, 3.5.2 and 3.6.5 when functioned.

3.6.5 Ballistic requirements. The initiators shall meet the requirements of 3.6.5.1, 3.6.5.2, 3.6.5.3 and 3.6.5.4 when fired in accordance with 4.5.8.

3.6.5.1 Ignition delay time. The initiator upper limit (UL) and lower limit (LL) for ignition delay time shall be as defined in Table III.

TABLE III. Ignition Delay Times

Initiator	MSI P/N	LL (seconds)	UL (seconds)
JAU-75/A	826852-03	0.400	0.600
JAU-76/A	826852-02	0.250	0.380
JAU-77/A	826852-04	0.562	0.938

3.6.5.2 Maximum pressure. The upper and lower pressure limits for maximum pressure (psig) when functioned at the specified temperatures shall be as follows in Table IV:

TABLE IV. Maximum Pressure Limits

Initiator	Temp/Range (F)	LL (psig)	UL (psig)	Duration (sec)
JAU-75/A	-65	500	1760	N/A
JAU-75/A	+200	625	1760	N/A
JAU-76/A	-65 to +200	1260	1760	N/A
JAU-77/A	-65 to +200	2000	3600	0.002

3.6.5.3 Shear pin actuation pressure. The upper and lower limits for the shear pin actuation pressure (psig) when fired over the temperature range –65 to +200°F shall be UL = 625 psig and LL = 400 psig (see 4.5.8) for all except the JAU-76/A whose limits are LL=400 psig, UL=600 psig over the temperature range of –65 to +200°F.

3.6.5.4 Duration. The JAU-77/A Initiator shall maintain pressure between 2000 psig and 3600 psig for a minimum duration of 0.002 seconds when fired into a52 inch length of aircraft hose per MIL-H-25579, size no. 4, into a 0.5 cubic-inch closed bomb (298AS100) per 547AS100.

3.7 Misfires. There shall be no misfires (see 4.6).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Production inspection (see 4.3)
c. Lot acceptance inspection (see 4.4)

4.2 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.3), a first article inspection sample of 39 initiators conforming to each drawing 3205AS266, 3205AS267 and 3205AS268 shall be subjected to first article testing as required. Thirty-nine of these initiators shall be expended in the tests listed in Table V. The Government shall retain any remaining units for investigative purposes. The tests shall be performed in accordance with MIL-D-21625. Any damage inflicted by the environmental exposures which would adversely affect the performance of the item in it's service application shall be cause for rejection of the first article sample. Any production prior to notification by the contracting agency of the first article sample acceptability shall be at the contractor's risk. Failure of any initiator to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

TABLE V. First Article Test Schedule

Test	Test method paragraph	Number of initiators					
		Group I	Group II	Group III	Group IV	Group V	
Visual inspection	4.5.1	18	3	6	6	6	39
Radiographic inspection	4.5.2	18	3	6	6	6	39
Temp.shock/humidity/altitude	4.5.4	18	-	-	-	-	18
Vibration	4.5.5	18	-	-	-	-	18
Shock	4.5.6	18	-	-	-	-	18
6 Foot Drop	4.5.7	-	3	-	-	-	3
Visual inspection	4.5.1	18	3	-	-	-	21
Radiographic inspection	4.5.2	18	3	-	-	-	21
JAU-76/A & JAU-77/A Initiators							
Ballistic testing at –65°F	4.5.8	6	-	6	-	-	12
Ballistic testing at +70°F	4.5.8	6	3	-	6	-	15
Ballistic testing at +200°F	4.5.8	6	-	-	-	6	12
JAU-75/A Initiator							
Ballistic testing at –65°F	4.5.8	9	2	3	3	3	20
Ballistic testing at +200°F	4.5.8	9	1	3	3	3	19

4.3 Production inspection. All production cartridges or initiators manufactured under the contract shall be inspected and screened for defects. Initiators failing to meet the requirements listed in Table II shall be rejected and removed from the lot. A production log providing a cross-reference of cartridge serial numbers to initiator serial numbers shall be maintained.

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4.4 Lot acceptance test inspection. Lot acceptance inspection shall consist of the examinations and tests specified in Table VI. Failure of any sample initiator to comply with the requirements listed in Table VI shall be cause for rejection of the lot represented. Packaging defects shall be corrected before acceptance.

4.4.1 Sample size. A random sample of initiators from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with Table VII for lot acceptance inspection. Test sample initiators and samples retained for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or purchase order.

TABLE VI. Lot acceptance inspection and tests.

Inspection/Test	Test Paragraph	Requirement Paragraph	Quantity
1. Visual Inspection	4.5.1	3.5.1	Test and retain sample
2. Radiographic Inspection	4.5.2	3.5.2	Test and retain sample
3. Leakage	4.5.3	3.5.3	100% at cartridge level
4. Packaging Inspection	N/A	Contract Section "D"	Test and retain sample
JAU-76/A & JAU-77/A Initiators Ballistic testing at –65°F Ballistic testing at +70°F Ballistic testing at +200°F	4.5.8	3.6	 1/3 sample 1/3 sample 1/3 sample
JAU-75/A Initiator Ballistic testing at –65°F Ballistic testing at +200°F	4.5.8	3.6	 1/2 sample * 1/2 sample

* If the sample is not equally divisible by two, the extra specimen shall be placed in the low temperature group.

TABLE VII. Ballistic sample size for given lot size.

Lot size	Sample size	Retained sample
1-50	8	3
51-90	13	3
91-150	20	3
151-280	32	3

4.5 Inspections and tests.

4.5.1 Visual inspections. The external condition and appearance of the initiators shall be determined by comparison to drawings 3205AS266, 3205AS267 and 3205AS268 as required.. All initiators and cartridge assemblies shall be visually examined and compared to the appropriate drawing for proper assembly, identification, color and interface control points. Those with illegible or inadequate descriptive markings or serial numbers, dents, deep scratches, or any other defects, which may prevent proper installation or

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functioning, shall be rejected. Any visual defects found during nondestructive inspection and destructive tests will result in 100 percent screening of the production lot at the contractor's expense. Each initiator shall meet the requirements of 3.5.1.

4.5.2 Radiographic examination. X-ray radiographic examination shall be conducted in accordance with ASTM E 1742, N-ray examination shall be in accordance with ASTM E 748. All initiators shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on boards or trays in consecutive order, with any missing serial numbers identified on the radiographic plate. The initiators shall be oriented such that the set screw and shear pins are visible on the radiograph. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4x6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests. Any observable imperfections as outlined in 3.5.2 shall be cause for rejection of the initiator. Defective initiators found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Settings on the radiographic equipment shall be noted on the radiographic review sheets.

4.5.3 Leakage test. The leakage test shall be conducted at the finished cartridge assembly level before installing external o-ring, and prior to assembly in the initiator assembly. The cartridges shall meet the requirement of 3.5.3. Production units shall be leak tested by flushing the loaded cartridge assembly with Helium at a pressure of 1+0.1/-0.0 atmospheres absolute. The cartridge assembly shall then be washed with dry Nitrogen and placed in a vacuum chamber and checked for leakage at one atmosphere differential. Certification of the test results shall be supplied and Government witnessing will be as specified in the contract or purchase order.

4.5.4 Temperature-shock/humidity/altitude (TS/H/A) cycling. The initiators with environmentally sealed protective caps and plugs installed on the inlet and outlet ports shall be submitted to the TS/H/A cycling test outlined in Table VIII with the following requirements:

a. The initiators shall be supported in such a way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No initiator shall be in contact with another initiator during the cycling schedule shown in Table VIII.

b. Failure of any initiator to comply with the requirements of 3.6.1 shall be cause for rejection of the first article.

c. There shall be no interim withdraws of units.

4.5.5 Vibration. This test is a combined sine on random environment and includes gunfire vibration. The vibration test shall be conducted per the following procedure. Failure of any initiator to comply with the requirements of 3.6.2 shall be cause for rejection of the First Article.

4.5.5.1 Non-Gunfire Random & Sinusoidal Requirements. The combined random and sinusoidal (non-gunfire) requirements involve a combination of sinusoidal cycling (5 to 2000 Hz) and high frequency (50 to 2000Hz) random vibration. The procedures shall be conducted along all three orthogonal axes at the following temperatures; -65F, +70°F, and +200°F.

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a. **Sinusoidal Vibration Test**. The initiators shall be vibrated at 70°F along the same orthogonal axis between 5 Hz to 50 Hz and returning to 5 Hz in approximately 7.5 minutes for a total time of 60 minutes per axis per Figure 1. Repeat at -65°F, and then again at +200°F.

b. **Random Vibration Test**. The initiators shall be vibrated along the same orthogonal axis at 70°F for 120 minutes in accordance with the applicable random vibration spectrum shape, test levels and time per Table IX and Figure 2. Repeat at -65°F, and then again at +200°F.

4.5.5.2 Gunfire Sinusoidal Vibration. This test is a supplemental vibration test for equipment located in regions of the aircraft affected by gunfire. In this procedure, a sinusoidal vibration test at +70°F is used in the frequency range of 50 to 2000 Hz. Testing shall consist of logarithmic sweeps from 50 to 2000 Hz and back to 50 Hz in 15 minutes plus fixed dwell periods per Tables VIII, IX, and Figure 3.

a. **Vibration Cycling.** The initiators shall be vibrated along the same orthogonal axis, with the frequency varying at a logarithmic rate from 50 to 2000 Hz and back to 50 Hz in approximately 15 minutes at double amplitude or vibratory acceleration levels in accordance with applicable gunfire test curves (see table X and Figure 3).

b. **Fixed Dwell.** The initiators shall be vibrated along the same orthogonal axis, at the fixed narrow band dwell frequencies. The fixed narrow band dwells shall be performed by sweeping the frequencies about ± 5% of the specified center frequency, see Table XI. The sweep rate shall be approximately the same as in the above paragraph. The vibration test levels shall be as shown in Figure 3.

4.5.6 Shock. The initiators with protective caps installed on the ports, shall be submitted to the shock test outlined in MIL-STD-810, Method 516.4 Procedure I to the levels in Figure 4, three (3) shocks in each direction along each of the three (3) axis for a total of eighteen (18) shocks. All shocks will be conducted at ambient temperature. Failure of any initiator to comply with the requirements of 3.6.3 shall be cause for rejection of the first article.

4.5.7 Six-Foot Drop Test. One initiator shall be dropped 6 feet (+2.0 inch, -0.0 inch) onto a 2.0-inch thick steel plate embedded into concrete. The drop orientation shall be one heads-up, one heads-down and one in a horizontal orientation. The initiators shall be dropped without shipping caps or plugs. The initiator shall meet the performance requirements of 3.6.4.

4.5.8 Functional performance. The initiators shall be test fired in the fixtures referenced in Table XII. Sample initiators for the functional performance test firings shall be conditioned for not less than 4 hours and no more than 24 hours at the specified temperature. The initiators shall be removed from the conditioning chamber and fired in the designated test fixture within 5 minutes. An initiation pressure of 800 +25/-0 psig shall be applied to the initiator inlet port at a pressure on-set rate of 33,000 ±500 psig/second. If any initiator is not fired within the 5 minute time period after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The initiator shall meet the performance requirements of 3.6.5.

4.6 Acceptance criteria. The initiators shall meet the requirements of 3.6 when tested as specified in 4.5.3 through 4.5.8. The acceptable number of defects is 0 and the rejection number is 1.

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TABLE VIII. Temperature shock/humidity/altitude cycling schedule

Monday	0800	Place initiators in chamber at 70°F at 50% relative humidity (RH).
	1200	Raise chamber temperature to 160°F and a RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the initiators from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi)
Tuesday	0800	Remove initiators from chamber and immediately place in chamber maintained at 70°F at 50% RH.
	1200	Remove the initiators from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi)
	1600	Remove Initiators from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Wednesday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and a RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the Initiators from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi)
Thursday	0800	Remove Initiators from chamber and immediately place in chamber maintained at 70°F at 50% RH
	1200	Remove the Initiators from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi)
	1600	Remove Initiators from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Friday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and a RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the Initiators from above chamber and immediately place in a chamber maintained at -65°F at standard ambient pressure.

This schedule shall be followed for a total of four (4) weeks (28 days) except that on the second and fourth weekend the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160°F at 95% RH.

TABLE IX. Random Test Levels

Frequency (Hz)	Endurance Amplitude (G**2/Hz) or Slope
10-670	0.04
670-1000	+ 5 dB/oct
1000-1300	0.078
1300-2000	-25 dB/oct

Note: Performance @ 5.2 Grms, Endurance @ 9.1 Grms

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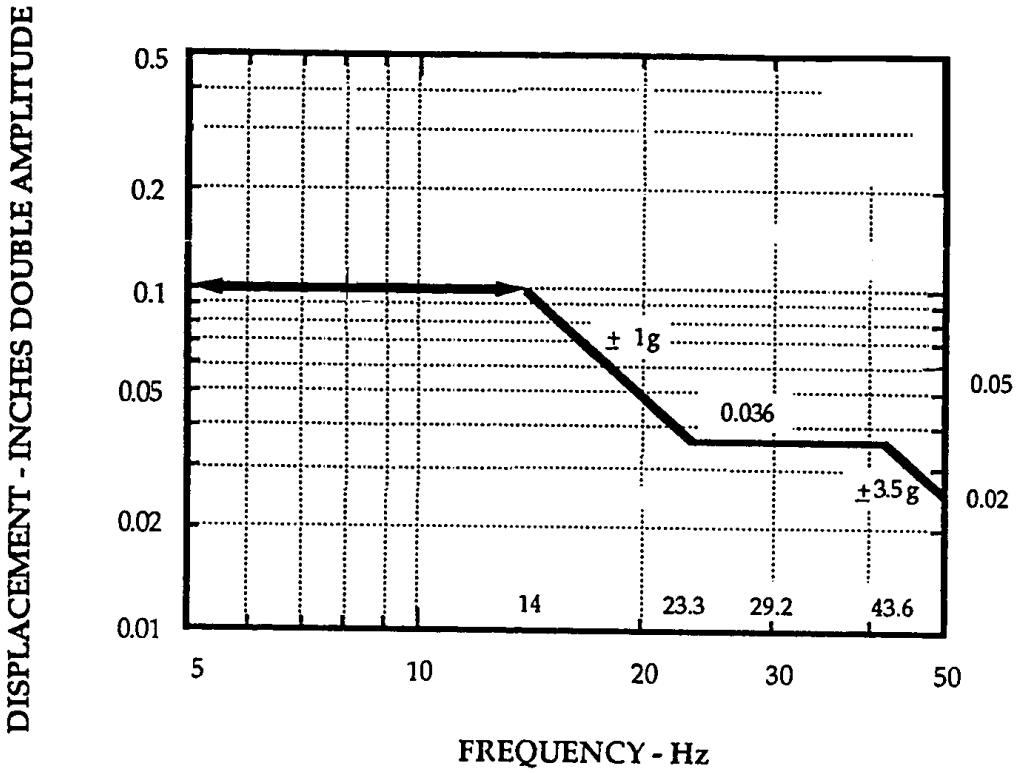


FIGURE 1. **Low Frequency Vibration Profile**

TABLE X. Vibration (Gunfire)- Sinusoidal Test Schedule

Sweep time (min/axis)	60
Fixed Dwell (f_d) at the following Center frequencies (see note 1)	
67 Hz (63.7-70.4 Hz)	7.5
100 Hz (95-105 Hz)	7.5
135 Hz (128-142 Hz)	7.5
200 Hz (190-210 Hz)	15.0
267 Hz (254-280 Hz)	7.5
300 Hz (285-315 Hz)	7.5
335 Hz (318-352 Hz)	7.5
400 Hz (380-420 Hz)	15.0
467 Hz (444-490 Hz)	7.5
500 Hz (475-525 Hz)	7.5
Total Time Each Axis (minutes)	150.0

Note 1. The fixed narrow band dwells shall be performed by sweeping the frequency about +/- 5% of the specified center frequency (f_d). The sweep rate shall be approximately the same as used for the sweep testing in the same range.
Note 2. For sinusoidal sweep tests, the frequency shall be varied at a logarithmic rate for the times indicated.

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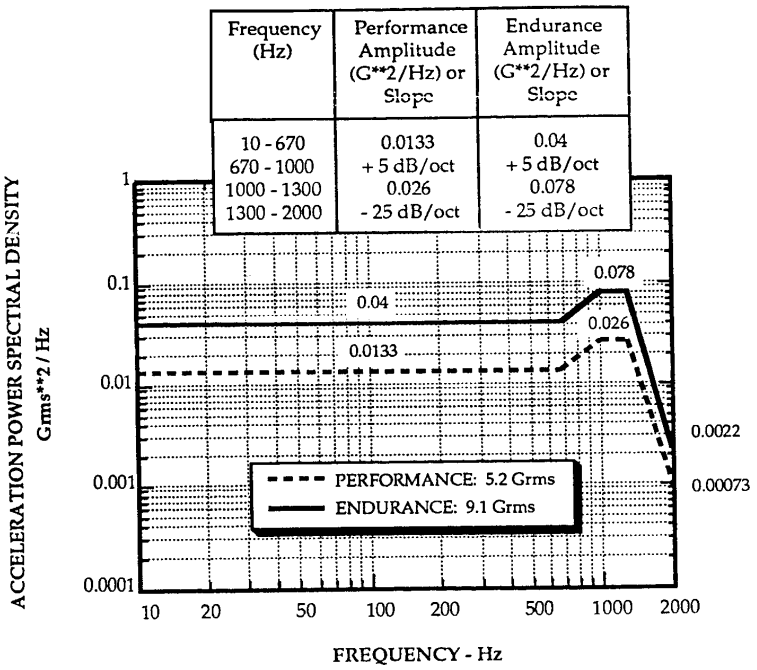


FIGURE 2. **Non-Gunfire Random Vibration Test Levels**

TABLE XI. Vibration Gunfire Sinusoidal Test Levels

Displacement-Inches Double Amplitude	Frequency - Hz
0.025	50
0.003	140
0.003	210
0.0015	210
0.00014	650
0.00014	1000
0.0000343	2000

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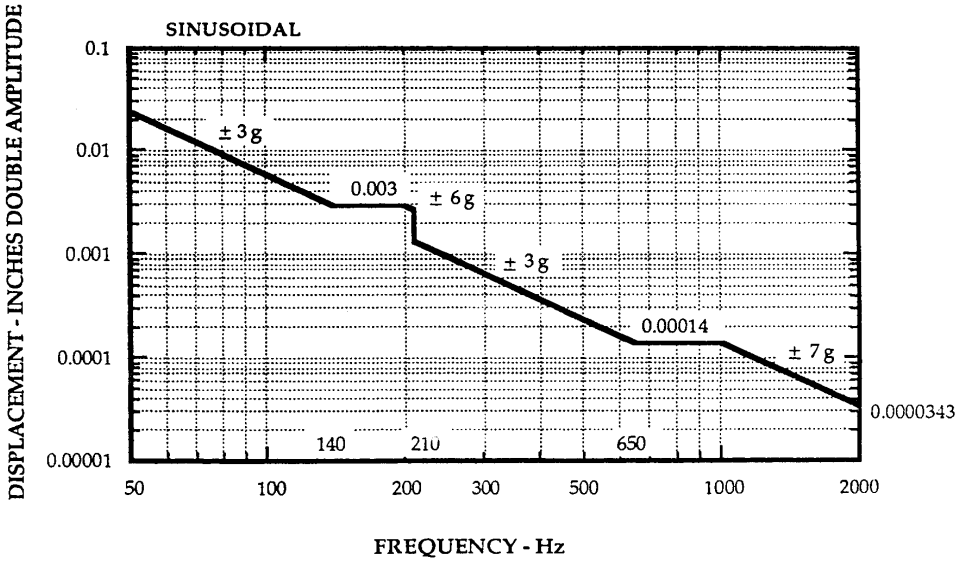


FIGURE 3. **Gunfire Sinusoidal Vibration Curve**

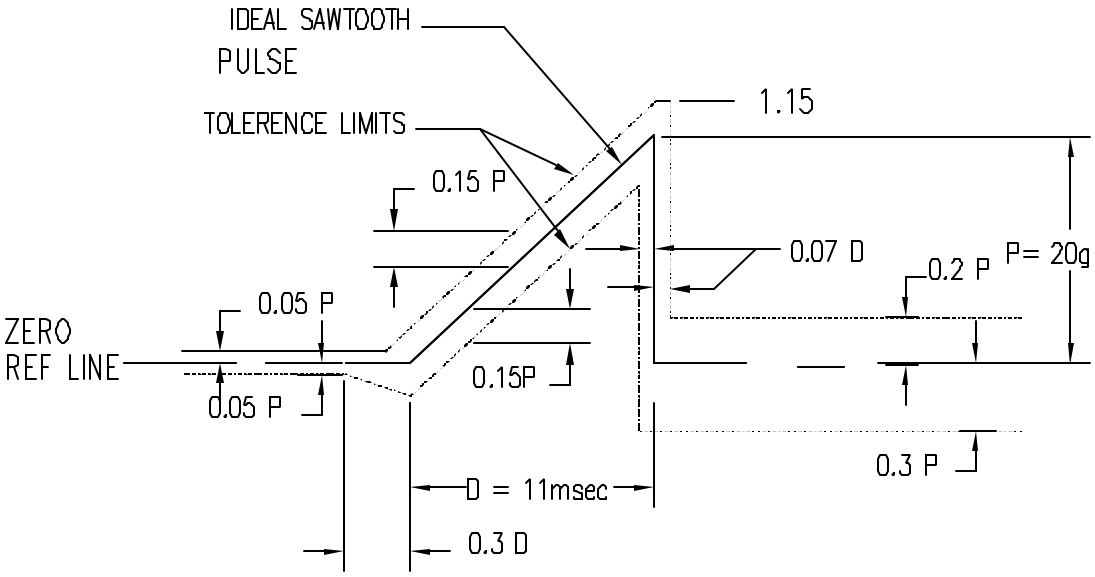


FIGURE 4. **Terminal peak saw tooth shock pulse**

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TABLE XII. Test firing fixtures.

Model	Hose Length (in)	Hose per	Hoze Size	Closed Bomb Volume (in^3)	Fixture Drawing CAGE 30003
JAU-77/A	52	MIL-H-25579	4	0.5	547AS100
JAU-76/A	N/A	N/A	N/A	5.0	726AS802
JAU-75/A	180	MIL-H-38360	4	0.5	848AS225

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended Use. These time delay initiators are intended for in the emergency escape systems of the S-3B, ES-3A, QF-4N/S, F-18A/B/C/D/E/F and T-45A/C Aircraft. These initiators were specifically designed for use in these military aircraft, and thus have no commercial application.

6.1.1 Changes to configuration. Changes to the initiators resulting in pressure and timing changes could result in catastrophic conditions to the crew during emergency escape. Any changes affecting the ballistic functions of these initiators require the approval of Naval Air Systems Command, AIR-4.1.1 and PM-201, and the prime airframe and ejection seat contractors.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).

c. Whether a first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2).

d. Markings if other than as specified in 5.1.

e. If bar coding is not required on outer container.

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f. Inspection conditions if other than specified in 4.2.

g. Production lot sizes (quantity) and test facility.

h. Configuration of DCA required (see 3.3.2).

i. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity the safety precautions requirements of "Ammunition and Explosives," OP 5, are applicable.

6.3 First Article. When a first article inspection is required, the contracting officer should provide specific guidance to offeror(s) whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Flight critical part. A part, the single failure of which, during any operating condition, could cause the loss of aircraft or one of its major components, loss of control, or which may cause significant personnel injury, in all phases of aircraft operations.

6.4.2 Primary component. Primary components are all components in which a functional failure would result in a misfire or malfunction of the gas generator.

6.4.3 Caps. A plastic plug should be installed in the inlet port and a non-propulsive cap installed on the outlet fitting of all initiators.

6.5 Contract packaging and markings. The following marking requirements should be specified in the contract.

6.5.1 Markings.

6.5.1.1 Special markings. Marking of interior, intermediate and exterior containers should be in accordance with 49 CFR 171-178 and MIL-STD-129.

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6.5.1.2 Normal marking. Unless otherwise specified in the contract or purchase order (see 6.2), the marking on the inner and outer containers should be in accordance with Special Packaging Instruction (SPI) No. 520-174-0283.

6.5.1.3 Bar Coding. Should be as specified in Section-D of the contract or purchase order (see 6.2), the outer container should be bar coded in accordance with MIL-HDBK-129. Additionally, the above-specified marking information should be applied to the containers in accordance with the provisions of MIL-HNDK-129.

6.5.1.4 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28B should be used to clarify terms.

6.6 Subject term (key word) listing.

Time-Delay
Propellant
Initiator

Preparing Activity:
Navy - OS
(Project 1377-0237)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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